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With USD 40+ million as seed capital within the first five years vision (Vision 2026-2031), AFRICAN GREEN FARMERS (AGF) Limited is a vertically integrated agro-industrial Zambian privately-owned agro-industrial biotechnology enterprise steadily engaged in transforming commercial organic farming of target and selective medicinal plants, functional crops, fruits and nutrient-dense agricultural resources into standardized plant-based TCAMs (Traditional, Complementary and Alternative Medicines), Nutraceuticals/Food Supplements, clean-label fortified health food products and beverages completely free from added unsafe sugar and chemicals, High-nutrient organic fertilizers as well as scientifically advanced formulation of phytomedicines and phytopharmaceuticals subjected to preclinical and clinical trials via our scientific research laboratory and laboratory animal breeding facility, as well as clean-label fortified health food products and fruit beverages. By integrating climate-smart commercial organic farming, structured out-grower networks, laboratory-driven product profiling, standardization preclinical & clinical trials, and scalable processing infrastructure, AGF shifts agriculture from low-margin primary commodity production to high-value industrial farming and manufacturing. The company is building a growth platform designed to expand cultivated acreage, increase processing capability, strengthen regional export penetration, and generate sustainable employment, especially of youth and women while contributing to Zambia's economic transformation and natural health value chain development.

We are targeting multi-million-dollar annual turnover, sustainable EBITDA performance, and scalable export expansion by 2031 (Vision 2026-2031).

In short:

AGF's facility (AGF-PHYTOMED INDUSTRIES) is engaged in manufacturing the following standard and GMP-compliant key products:

- A) CAMs (Traditional, Complementary and Alternative Medicines), targeting acute and chronic diseases;
- B) Nutraceuticals/Food Supplements targeting acute and chronic diseases;
- C) Clean-label Fortified Health Food Products & Beverages for prevention and fight against malnutrition especially among children, pregnant women breast feeding mothers;
- D) High-nutrient organic fertilizers for soil regeneration and enrichment;
- E) Phytomedicines, targeting acute and chronic diseases;
- F) Phytopharmaceuticals, targeting acute and chronic diseases.



TAKE ACTION

Investors & Donors: To submit your **Manifestation of Interest (MOI)**

International Brokers: Apply for Strategic Brokerage Agreement

Buyers: Request Product Catalogue (GMP/ISO compliant)

Patients: Book Telehealth Consultation

Page2: About Us

1) Project Fact Sheet

Project Title: AGF-PHYTOMED INDUSTRIES Integrated Phytopharmaceutical & Nutraceutical Manufacturing facility

Promoter entity: African Green Farmers Ltd (AGF)

Location:

Registered Address:

Mobile:

Emails:

Website:**Sector:** Pharmaceutical Manufacturing & Botanical Standardization**Total Seed Capital Requirement:** USD 40+ Million**2) Corporate Profile**

- Company commercial name: African Green Farmers (AGF) Limited
- Fully incorporated limited liability Company on 12th March 2025
- TPIN number: 2003675243`
- Investment License secured
- Sector: Agro-Processing & Phytomedicine/phytopharmaceutical Manufacturing
- Ownership Structure: Privately held by founding partners

3) Brief Company Overview

AFRICAN GREEN FARMERS (AGF) Limited is a vertically integrated agro-industrial Zambian privately-owned company engaged in commercial organic farming of key medicinal plants, functional crops, fruits and vegetables in over 2000 hectare-farmland and industrial processing of innovative standardized TCAMs (Traditional, Complementary and Alternative Medicines), Nutraceuticals/Food Supplements, scientifically advanced formulation of phytomedicines and phytopharmaceuticals subjected to preclinical and clinical trials via our scientific research laboratory and laboratory animal breeding facility, as well as clean-label fortified health food products and fruit beverages. Through advanced value addition, scientific quality control & standardization and inclusive farmer partnerships, AGF transforms organic agriculture into scalable industrial opportunity and sustainable economic growth via our AGF-PHYTOMED INDUSTRIES facility.



4) Core Business Activities

The following are key activities:

- 1) Corporate documentation
- 2) Business plan development
- 3) Fundraising
- 4) Strategic partnership development
- 5) Regulatory affairs documentation
- 6) Licensing and compliance documentation
- 7) Corporate governance documentation
- 8) Investment documentation
- 9) Statutory documentation
- 10) Development finance documentation
- 11) Regulatory submission
- 12) Permit and certification documentation
- 13) Corporate structuring documentation
- 14) Compliance and authorization documentation
- 15) Business regulatory advisory
- 16) Vendor management
- 17) Progressive employee recruitment
- 18) Turnkey construction of the AGF-PHYTOMED INDUSTRIES facility
- 19) Importation and commissioning
- 20) Product manufacturing, certification, registration, Marketing and market launch
- 21) Agricultural land sourcing and organic farm development

5) Vision Statement

To become a leading Zambian-based model in chemical-devoid large-scale farming of organic crops and medicinal plants and industrial hub for manufacturing of innovative standardized TCAMs, Nutraceuticals/Food Supplements, phytomedicines and phytopharmaceuticals subjected to preclinical and clinical trials via our scientific research laboratory and laboratory animal breeding facility, high-nutrient organic fertilizers as well as clean-label fortified health food products and fruit beverages- recognized across African continent and globally for quality, safety and sustainability, and contribution to the sustainable socio-economic transformation.

6) Mission Statement

To organically do large-scale commercial farming of key medicinal plants, functional crops, fruits and vegetables in over 2000 hectare-farmland and to process the harvest into GMP-compliant and ISO standard TCAMs, Nutraceuticals/Food Supplements, phytomedicines and phytopharmaceuticals subjected to preclinical and clinical trials via our scientific research laboratory and laboratory animal breeding facility, as well as clean-label fortified health food products and fruit beverages and high-nutrient organic fertilizers.

7) Strategic Positioning

AFRICAN GREEN FARMERS LIMITED is positioned as a vertically integrated agro-industrial company focused on high-value medicinal and functional crops. Through structured partnerships, value addition and export-oriented processing, the company transforms commercial organic agriculture into scalable industrial (TCAMs, Nutraceuticals/Food Supplements, phytomedicines and phytopharmaceuticals subjected to preclinical and clinical trials via our scientific research laboratory and laboratory animal breeding facility, as well as clean-label fortified health food products and fruit beverages.

8) OBJECTIVES

8.1. Global Objective

Following the strategic deployment of USD 40+ million as seeds capital, AFRICAN GREEN FARMERS (AGF) Limited will have fully constructed and operationalized the AGF-PHYTOMED INDUSTRIES facility as a GMP-aligned agro-biotechnology manufacturing platform producing essentially seven (7) categories of standardized TCAMs, Nutraceuticals/Food Supplements, clean-label fortified health food products and fruit beverages, high-nutrient organic fertilizers, preclinical research-based phytomedicines and phytopharmaceuticals, generating sustainable job opportunities and revenues from domestic markets while achieving ISO certifications, entitling us for regional and international export expansion, targeting multi-million-dollar annual turnover, sustainable EBITDA performance, and scalable export expansion by 2031 (Vision 2026-2031).

9.2. Specific objectives

- (1) Establish a GMP-compliant AGF-PHYTOMED INDUSTRIES facility endowed with ISO-certified cleanroom manufacturing standards
- (2) Develop standardized TCAMs, Nutraceuticals/Food Supplements, phytochemicals & phytopharmaceuticals formulations, fortified clean-label health food products and fruit beverages
- (3) Integrate fermentation biotechnology for high-nutrient organic fertilizers
- (4) Setting Analytical and Scientific Research Laboratory, as well as featuring advanced analytical and research tools, including biotechnology-grade instruments such as:
 - Ultra-high Performance Liquid Chromatography (UHPLC-MS/MS)
 - Triple Quadruple Gas Chromatography Tandem Mass Spectrometry (GC-MS/MS 8000)
 - High-Performance Atomic Absorption Spectrometer (HPAAS)
 - AH SN600 Laser Scanning Microscope, Quant Studio 5 (RT-qPCR and
 - High-Parameter Digital Flow Cytometer that are strongly recommended for product profiling and standardization, as well as preclinical and clinical trials
- (5) Setup preclinical animal research facility where laboratory animals (e. g. Mice) are housed, bred and used under controlled scientific research conditions for generating scientifically validated, regulatory compliant preclinical data that proves a product is safe, effective, and standardized before human administration
- (6) Undertake commercial-scale organic cultivation of medicinal plants, functional crops, fruits and vegetables for sustainable raw material production
- (7) Achieve domestic, regional and international regulatory approvals
- (8) Scale export-oriented manufacturing for domestic, regional and international market integration

10. Specific Strategic Objectives (2026–2031)

10.1. Commercial Organic Farming Expansion Objective

Scale climate-smart commercial organic cultivation of high-value medicinal and functional crops.

Phase I (2026–2027)

- Operationalize 90 hectares already acquired.
 - ✓ Cultivate priority medicinal plants including *Carica papayaya* and *aloe vera*
 - ✓ Establish irrigation, post-harvest handling, and drying systems.

Phase II (2028–2031)

- Expand to 2,000+ hectares (owned + contracted farming).
- Integrate 5,000+ out-growers.

- Achieve certified organic compliance (regional/international standards).
- Implement digital traceability across $\geq 70\%$ of cultivated area.
- Target Output by 2031:

15,000–25,000 metric tons annual raw biomass production.

10.2. Advanced Processing & Manufacturing Infrastructure Objective

Deploy USD 40+ million seed capital into advanced agro-biotechnology infrastructure.

Capital Allocation Areas:

A. Facility Establishment

Investment in:

- 1) Construction of GMP-grade facility (multi-zone design) and commissioning.
- 2) ISO-classified cleanroom establishment essentially for:
 - Post-harvest processing, extraction, concentration, chromatographic purification and drying of plant-based bioactive compounds
 - Encapsulation and tableting
 - Bottling, bottle sealing and labeling, profiling and standardization, cartooning and case packaging

B, Manufacturing Line Systems

Investment in:

- 1) A fully integrated and automated Turnkey Ultrasound-Assisted Extraction (UAE), Capsule Production, Tableting and packaging Line System customized to manufacture:
 - TCAMs (Traditional, Complementary, and Alternative Medicines)
 - Nutraceuticals/Food Supplements
 - Phytomedicines
 - Phytopharmaceuticals
- 2) A fully integrated and automated Turnkey Line System designed specifically for:
 - Fortified Liquid Soybean Milk
 - Instant Fortified Soymilk Powder
 - Fortified Vacuum Dried Tofu
- 3) A fully integrated and automated Turnkey Line System designed specifically for spirulina algae production and harvesting, powder, capsule and tablet processing as well and packaging:

- Fortified & Micronized Spirulina Algae Powder (Food Supplement)
 - Fortified Spirulina Algae Capsules (Food Supplement)
 - Fortified Spirulina Algae Tablets (Food Supplement)
- 4) A fully integrated and automated Turnkey Cold Press Line System designed specifically for juice processing and packaging:
- Fortified Cold-pressed Fruit Juices
- 5) A Fully Integrated and Automated Processing and Manufacturing Line System for High-Nutrient Organic Fertilizers:
- Liquid concentrates (Via vacuum evaporation) organic fertilizer
 - Micronized Powder

C. Advanced Laboratory Equipment

Investment essentially in:

Analytical and Scientific Research Laboratory, featuring advanced analytical and research tools, including biotechnology-grade instruments, such as the:

- Ultra-high Performance Liquid Chromatography (UHPLC-MS/MS)
- Triple Quadruple Gas Chromatography Tandem Mass Spectrometry (GC-MS/MS 8000)
- High-Performance Atomic Absorption Spectrometer (HPAAS)
- AH SN600 Laser Scanning Microscope
- Quant Studio 5 (RT-qPCR)
- High-Parameter Digital Flow Cytometer

For: product profiling and standardization for quality control, safety and efficacy, as well as preclinical and clinical trials ➤ Regional and international market credibility.

10.3. Product Portfolio Development Objective By 2031, commercialize:

A. Standardized TCAMAs

TCAMAs (Traditional Complementary & Alternative Medicine Adaptations), 20+ standardized botanical formulations, Stability-tested and batch-traceable.

B. Nutraceuticals & Food Supplements

15+ evidence-supported supplements, Clean-label, non-synthetic fortified products.

C. Fortified Functional Foods & Beverages

Nutrient-dense fruit & vegetable derivatives, Botanical-enhanced health beverages.

D. Clinical Trial–Based Phytomedicines & Phytopharmaceuticals

Pre-clinical validation via animal research facility, at least 3 clinical trial programs initiated by 2030.

Target acute and chronic conditions (metabolic, inflammatory, immune-related).

10.4. Quality, GMP & ISO Certification Objective By 2029:

- ✓ Achieve ISO 9001 certification.
- ✓ Achieve ISO 22000 certification.
- ✓ Implement full GMP documentation & SOP system.
- ✓ Establish pharmacovigilance and post-market surveillance.
- ✓ Secure regulatory approvals for domestic commercialization.
- ✓ Prepare export dossiers for regional markets (COMESA/SADC).

This ensures readiness for international pharmaceutical-grade market entry.

10.5. Financial & Market Performance Objective By 2031:

- ✓ Annual turnover: USD 15–25 million
- ✓ Gross margin $\geq 45\%$
- ✓ EBITDA $\geq 22\%$
- ✓ Positive cash flow by Year 3 post-operationalization
- ✓ Export share $\geq 30\%$ of total revenue.

10.6. Socio-Economic & ESG Impact Objective

- ✓ 500+ direct skilled industrial jobs.
- ✓ 5,000–10,000 indirect rural livelihoods.
- ✓ $\geq 60\%$ youth and women participation.
- ✓ 30–50% farmer income increase.
- ✓ Regenerative agriculture improving soil carbon content $\geq 20\%$.

Clarification on USD 40+ Million Seed Capital Use

The USD 40+ million seed capital will be allocated strategically as follows:

- Estimated Allocation Range:
 - GMP-compliant Facility Construction: 25–35%
 - Advanced Lab Equipment: 15–20%
 - Processing & Manufacturing Lines: 20–30%
 - Cleanrooms & Sterile Zones: 8–12%

- Certification & Regulatory Compliance: 3–5%
- Working Capital Buffer: 10–15%

This level of capitalization positions AGF not as a small herbal company- but as a regional agro-biotechnology and phytopharmaceutical industrial platform.

▪ Strategic Positioning Shift

With USD 40M+ capitalization, AGF transitions from:

- ✓ Initial agro-processing → large-scale processing
- ✓ Regional phytopharmaceutical industrial platform.

11. Our Vision 2026-2031 (Milestones)

1) Research & Product Development

- 2026: Complete bioactive compound identification for top 5 chronic disease targets (oncology, diabetes, cardiovascular).
- 2027: Preclinical validation (mechanism-of-action, safety, efficacy) of lead compounds.
- 2028: Submission of first regulatory dossiers for clinical trials.

2) Clinical & Healthcare Services

- 2026–2027: Launch pilot on-site herbal clinic in capital city+online health consultation platform (phone, form-based, video) for domestic and regional coverage.
- 2027–2028: Expand to 5 additional domestic clinics.
- 2028–2030: Establish “AGF ONE” AI-assisted platform (Application) to provide a global AI-powered platform for learning, health, marketplace, and community engagement, while enabling scalable revenue tracking and growth
- 2029: Begin first clinical trial in collaboration with regional or international hospital networks.

3) Community Training & Capacity Building

- 2026: Develop community training curriculum for commercial organic farming of medicinal plants.
- 2027: Train first 500 community members in medicinal plant cultivation and post-harvest processing.
- 2028–2030: Scale to train 5,000 community members across Zambia and COMESA region.

4) Preventive Nutrition & Health Education

- 2026: Launch multi-channel awareness campaigns (TV, radio, social media) on preventive nutrition and home remedies.
- 2027–2028: Reach 50,000 community members via campaigns and workshops.
- 2029–2031: Scale awareness programs across Southern Africa with digital platforms (WhatsApp, YouTube, Facebook, TikTok).

5) Manufacturing & Quality Assurance

- 2026: Construct and equip GMP-aligned phytopharmaceutical production facility.
- 2027: Obtain GMP and GLP certifications.
- 2028: Begin commercial production of standardized herbal medicines.
- 2029–2031: Achieve batch-to-batch consistency, full traceability, and supply to domestic and international markets.

6) Partnerships & Funding

- 2026–2027: Sign domestic and regional partnerships for research, training, and clinical networks.
- 2027–2028: Establish key international scientific and technical partnerships (universities, GLP labs, clinical networks).
- 2028–2031: Secure blended finance and institutional impact funding (DFIs, IFC, Gates Foundation).

7) Impact & Scaling

- 2028: Reach first 50,000 patients/clients with herbal and digital health services.
- 2029–2031: Expand ecosystem to cover multiple chronic diseases across Zambia, Southern Africa, and select international markets.
- 2031: AGF fully recognized as a regional leader in standardized phytopharmaceuticals and community health empowerment, with measurable social and health impact metrics.

8) Summary:

These milestones translate AGF’s objectives into concrete, measurable checkpoints, covering research, clinical services, community training, manufacturing, partnerships, and impact, making progress visible to investors, donors, and regulators.

Page3: Background & Strategic Rationale

Zambia, like many African nations, is endowed with a rich biodiversity of medicinal plants and a strong cultural heritage of traditional medicine use. However, the Traditional, Complementary and Alternative Medicines (TCAMs) sector remains largely underdeveloped, characterized by rudimentary practices, lack of standardization, minimal research support, and the absence of modern manufacturing infrastructure. Despite growing

political will, recognition by the Ministry of Health, and the establishment of regulatory bodies such as ZAMRA, the country still lacks the necessary technological capacity to transform traditional herbal knowledge into high-quality, safe, and standardized health products.

African Green Farmers Limited is structured, via its AGF-PHYTOMED Industries Facility, as a vertically integrated, GMP-aligned and ISO standard agro-biotechnology and phytopharmaceutical manufacturing platform designed to capture high-growth demand for plant-based preventive and therapeutic health products across domestic and regional markets. By scaling commercial organic cultivation from 90 hectares to over 2,000 hectares of high-value medicinal and nutrient-dense crops, including Aloe vera and Carica papaya, and deploying USD 40+ million into advanced processing infrastructure, cleanroom manufacturing environments, phytochemical and microbiological laboratories, and ISO-certified quality systems, AGF secures raw material traceability, establishes strong barriers to entry, optimizes value addition margins, and reduces regional import dependency. The diversified product architecture—spanning standardized TCAMs, nutraceuticals, clean-label fortified functional foods, high-nutrient organic fertilizers, clinically validated phytomedicines and phytopharmaceuticals—creates layered revenue streams with early cash-flow generation and long-term high-margin phytopharmaceutical positioning, targeting multi-million-dollar annual turnover, sustainable EBITDA performance, and scalable export expansion by 2031.

AGF-PHYTOMED INDUSTRIES facility emerges as one of leading Zambian private initiatives to implement state-of-the-art, fully integrated and automated industrial solutions in the herbal and nutraceutical sector. The company has adopted rare customized and automated Advanced GMP-Compliant and ISO- standards Turnkey Production Complex for Herbal API (**Active Pharmaceutical Ingredients**) Extraction, Encapsulation and Tableting line system. This complex industrial-scale and automated setup “*TURNKEY ULTRASOUND-ASSISTED EXTRACTION (UAE), CAPSULE PRODUCTION AND TABLETING LINE SYSTEM*” combines three advanced processing techniques:

- 1) Phytochemical Extraction and Purification Process- Post-harvest processing, eventual defatting, concentration, eventual purification. vacuum drying and packaging the extract for storage;
- 2) Capsule Manufacturing Process- Encapsulation, bottle filling, capping and sealing, bottle instructional labeling, bottle cartoning, leaflet inserting and sealing and case packaging of 15 bottles;
- 3) Tablet Manufacturing Process- Dry extract blending, granulation, eventual vacuum drying, fluid bed dryer, tablet compression, tablet coating, bottle filing, capping, bottle

sealing, bottle instructional labeling, bottle cartoning, instructional leaflet inserting as well as the case packaging of 15 bottles.

The line system operates at reduced temperatures (40–60°C) to preserve heat-sensitive active pharmaceutical ingredients (Bioactive compounds) targeted to develop biologically safe and effective herbal medicines, Nutraceuticals, nutrient-dense fortified health food products and high-nutrient organic fertilizer—unprecedented cutting-edge technology in Zambia’s TCAM/phytomedicine sector.

Page 4: ESG & Sustainability

(1) Governance, Compliance & Transparency

AGF is committed to strong corporate governance, regulatory compliance, and financial transparency as foundational pillars of sustainable industrial growth. The USD 40+ million capital deployment as seed capital prioritizes GMP (Good Manufacturing Practice)-compliant infrastructure, ISO certification systems (ISO 9001, 22000, 14001, 45001 and 14644), maintenance of annual independent financial audits, ESG (Environmental, Social and Governance) performance reporting, risk management oversight, and board-level governance structures, ensuring accountability and institutional integrity.

Governance Commitments:

- ✓ Full GMP implementation and documentation
- ✓ ISO certification roadmap completion
- ✓ Annual independent financial audits
- ✓ ESG performance monitoring and reporting
- ✓ Structured compliance and risk management systems

(2) Institutional Oversight and Accountability

AGF Company operates under a structured corporate governance model that assures;

- Clear separation between shareholders, Boards, and Executive Management
- Defined fiduciary responsibilities and delegated authority frameworks
- Independent advisory expertise in regulatory science, finance, ESG and risk management
- Structured board oversight on strategy, compliance and capital allocation

This governance framework aligns with international best practices recognized by institutions such as International Finance Corporation and the World Bank.

(2) Regulatory Compliance, Quality Assurance & Licensing

AGF is building an integrated compliance ecosystem aligned with international recognized standards, including but not limited to:

- ISO 9001-Quality Management System
- ISO 22000-Food Safety Management
- ISO 14001- Environmental Governance and Sustainability
- ISO 45001- Occupational Health & Safety
- ISO 14644- Cleanroom Qualification

Our production roadmap integrates Good Manufacturing Practices (GMP), validated laboratory systems, traceability controls, and regulatory documentation protocols required for export-oriented TCAM, Nutraceutical/Food Supplement, Phytomedicine and Phytopharmaceutical, Fortified Clean-Label Health Food Product and Fruit Beverage regional and international markets.

AGF-PHYTOMED INDUSTRIES is subjected to the following regulatory licensing:

- Business Application & Registration –PACRA
- Investment Licence- ZDA
- Environmental Impact Assessment (EIA) Certificate- ZEMA
- Screening EPB Report Certificate- ZEMA
- Waste Management Licence- ZEMA
- Site Approval Certificate- ZMRA
- Product Registration- ZMRA
- GMP Inspection- ZAMRA
- Manufacturing Licence- ZAMRA
- Investor Permit- IMMIGRATION
- Product Testing & Application- ZABS
- Factory Inspection- ZABS
- Certification Mark- ZABS
- Premises Hygiene Clearance Certificate- Local Health Department (Munispal/District Health Office)
- Food Safety Inspection/Compliance Certificate- Local Health Department (Munispal /District Health Office)
- Food Handling/Safety Certificate- Local Health Department (Munispal /District Health Office)
- Occupational Health Risk Assessment Certificate- Local Health Department (Munispal/ District

(3) Transparency & Financial Integrity

We are committed to:

- Independent statutory audits
- Transparency financial reporting
- ESG performance disclosure
- Use-of-funds traceability for institutional capital
- Impact measurement and reporting aligned with sustainability metrics

AGF maintains zero unethical tolerance to fraud, bribery, unethical procurement, or regulatory non-compliance.

(4) Investor & Donor Assurance

Our governance and compliance systems are designed to ensure that capital deployed into AGF-PHYTOMED INDUSTRIES facility:

- Protected through structured oversight
- Allocated according to approved strategic milestones
- Monitored through measurable ESG and financial performance indicators
- Reported transparency to shareholders

(5) Strategic Investment Positioning Statement

AGF-PHYTOMED INDUSTRIES facility is being structured by African Green Farmers Limited as a bankable, audit-ready, ESG-aligned industrial platform capable of absorbing institutional capital responsibility while delivering measurable environmental, social, and financial returns.

Our governance framework reduces systemic risk, strengthens regulatory acceptance, and enhances international market credibility- positioning AGF as a trusted partner for long-term, impact-oriented investment.

Page 5: Research & Innovation

At African Green Farmers Limited, research and innovation constitute the scientific backbone of our vertically integrated model—from sustainable cultivation of medicinal plants and functional crops to the development of standardized TCAMs, Nutraceuticals/Food Supplements, clean-label fortified health food products and fruit beverages, preclinical research-based phytomedicines and phytopharmaceuticals, clinically validated phytopharmaceutical products for domestic, regional and international markets. Our research strategy integrates ethnobotanical intelligence, modern pharmacognosy, phytochemistry, molecular biology, biotechnology and translational preclinical sciences to transform indigenous bioresources into evidence-based, regulatory-compliant health products.

1. Integrated Research Architecture

Our innovation ecosystem is structured across five technical pillars:

B. Ethnobotanical Intelligence & Biodiversity Valorization

We systematically document, validate and prioritize medicinal plant species using evidence-based screening criteria, focusing on high-burden diseases, metabolic disorders, immune health and preventive nutrition. This ensures both scientific relevance and strong market viability.

B. Phytochemical Standardization & Molecular Profiling

Using advanced analytical platforms (UHPLC-MS/MS, GC-MS/MS 8000, HPAAS, AH SN600 Laser Scanning Microscope, RT-qPCR, and High-Parameter Digital Flow Cytometer), we:

- Identify active marker compounds
- Establish quantitative phytochemical fingerprints
- Ensure batch-to-batch reproducibility
- Develop standardized extracts meeting international pharmacopeial thresholds

This scientific rigor differentiates our products from traditional herbal preparations and positions them as standardized phytopharmaceutical-grade formulations.

C. Bioprocessing & Bio-enhancement Technologies

We apply controlled fermentation systems, microbial biotechnology (including selected LAB strains where applicable), and green extraction technologies to:

- Increase bioavailability
- Enhance active compound concentration
- Improve stability and shelf life
- Optimize pharmacodynamic performance

D. Preclinical & Translational Research

Through our planned laboratory animal research platform, we conduct:

- Toxicological profiling
- Pharmacokinetic and pharmacodynamic studies
- Dose optimization
- Mechanism-of-action validation

This bridges traditional knowledge with evidence-based medicine and supports regulatory approval pathways.

E. Product Development & Regulatory Alignment

We design formulations aligned with:

- WHO Traditional Medicine Strategy
- African Medicines Agency harmonization frameworks
- International GMP standards
- ESG and sustainability compliance benchmarks
- Each product progresses through defined Technology Readiness Levels (TRLs), ensuring clear de-risking milestones for investors.

F. Research-to-Market Pipeline

Our innovation pipeline follows a structured 7-stage model:

- Biodiversity mapping & species selection
- Phytochemical screening & active compound validation
- Standardized extract development
- Preclinical safety & efficacy studies
- Pilot-scale GMP production
- Regulatory dossier preparation
- Commercial scale-up & market deployment

This pipeline reduces technical uncertainty, accelerates time-to-market, and protects investor capital through phased risk mitigation.

G. Innovation with Measurable Outcomes (2031 Targets)

By 2031, AGF-PHYTOMED INDUSTRIES will have:

- ✓ Developed multiple standardized phytopharmaceutical product lines
- ✓ Established a fully operational research-grade analytical laboratory
- ✓ Filed proprietary formulations and intellectual property assets
- ✓ Achieved international-quality certifications
- ✓ Generated validated clinical data supporting global commercialization
- ✓ Positioned as a regional hub for evidence-based phytomedicine innovation

H. ESG-Driven Scientific Innovation

Our research platform is built on:

- Sustainable raw material sourcing
- Climate-smart organic cultivation
- Zero-waste bioprocessing
- Community-inclusive value chains
- Gender-inclusive scientific workforce development
- Scientific excellence and sustainability are not parallel strategies—they are integrated drivers of long-term value creation.

I. Sustainability & Long-Term Value Creation

AGF integrates climate-smart agriculture, soil regeneration, responsible resource use, and inclusive economic participation into its core operations. Expansion to over 2,000 hectares will integrate thousands of smallholder farmers, increase rural incomes, and generate skilled industrial employment while contributing to regional pharmaceutical self-reliance and sustainable industrial growth.

Page 6: Corporate Structure & Governance

AGF-PHYTOMED Industries operates under a competency-based corporate governance structure and Science-Driven Executive Leadership designed to ensure transparency, regulatory compliance, and sustainable industrial growth. The company is overseen by a professionally constituted Board of Directors providing strategic oversight, risk supervision, and investor protection. Executive management is structured across specialized functions including operations, scientific research, finance, quality assurance, and commercial development. Operational activities are organized into distinct divisions covering commercial organic farming, advanced processing and manufacturing, research and innovation, regulatory compliance, and corporate services. Independent audit, compliance, and ESG oversight mechanisms are embedded within the governance framework to ensure accountability, institutional integrity, and long-term enterprise resilience.

1) Administrative Structure

AGF will adopt a structured management framework with qualified professionals in key roles:

NO	POSITION	QUALIFICATION	MAJOR ROLE (S)
1	Chief Executive Officer (CEO)		Overall management and decision-making
2	Board of Directors		Provision of strategic direction
3	Chief Operations Officer-(COO)	PhD in Clinical pharmacology, pharmacokinetics or drug development	Overseeing production, quality control, R&D coordination, regulatory compliance, technical stuff, supply chain, facility operations, and safety
	Finance & Administration Manager	Master's degree in Business Administration or related field	Management of accounting and human resources
4	Director of Quality Control	Master's degree in Chemistry/Biotechnology/Food Science and Technology	Overseeing of correct and efficient manufacturing-related operations
5	Director of Industrial Maintenance	Master's degree in industrial maintenance, skilled in industrial automation, Good Manufacturing Practice (GMP)	Overseeing of installation, repair and maintenance of all facility's processing and research laboratory equipment

		and Standard Operating Procedures (SOPs)	
6	Head of (research) Laboratory	PhD degree in molecular biology/Biochemistry	Overseeing lab research direction, staff supervision and lab operations compliance
7	Sales and Marketing Manager	Bachelor degree in Marketing, Business Administration	Management of sales team, marketing stuff and distributors
8	Digital Marketing and Creative Design Manager	Bachelor degree in Marketing, communications and graphic design	Graphic design and mass media advertising
9	Supply and Logistic Officer	Bachelor degree in Logistics, Supply Chain Management	Overseeing procurement control, inventory and distribution
10	Corporate Lawyer	Holding Law degree/Corporate Business Law	Overseeing all company's legal matters
11	Director of Agricultural Operations	Master's degree in agricultural engineering	Overseeing the crop/medicinal plant organic industrial farming

2) Governance Committees & Personalities

2.1. Executive Management Committee (EMC)

- ✓ Strategic planning & policy approval
- ✓ Oversees departmental coordination
- ✓ Budget & investment decisions

2.2. Governance & Compliance

- ✓ Ensures regulatory compliance with national and international standards (e. g ZAMRA, ZEMA, GMP, ISO)
- ✓ Oversees ethical, legal and risk management frameworks across departments
- ✓ Monitors internal policies and guides organizational accountabilities and transparency

2.3. Quality Management Committee (QMC)

- ✓ Ensures GMP/ISO compliance
- ✓ Reviews internal audits, CAPA, deviations
- ✓ Approves changes to SOPs/QA systems

24. Ethics & Compliance Committee

- ✓ Monitors regulatory, legal, and environmental compliance
- ✓ Prevents conflicts of interest
- ✓ Promotes ethical business conduct

2.5. Product Development/R&D Committee

- ✓ Approves new formulations
- ✓ Oversees clinical validation (if any)
- ✓ Guides innovation strategy

2.6. Health & Safety Committee

- ✓ Implements occupational health/safety policies
- ✓ Risk assessment & emergency preparedness
- ✓ Coordinates safety training

3, Departments & Responsibilities

3.1. Production Department

- ✓ Executes manufacturing (extraction, formulation, packaging)
- ✓ Maintains batch records, GMP hygiene
- ✓ Manages equipment operations

3.2. Quality Assurance (QA)

- ✓ GMP implementation & document control (SOPs, CAPA)
- ✓ In-process and post-production checks
- ✓ Batch release authorization

3.3. Quality Control (QC)

- ✓ Lab testing (raw materials, in-process, finished products)
- ✓ Microbiology, chemical, and instrumental testing
- ✓ Stability testing and sampling

3.4. Research & Development (R&D)

- ✓ Product formulation & standardization
- ✓ Pilot production and validation
- ✓ Ingredient functionality research

3.5. Regulatory Affairs

- ✓ Handles product registration (ZAMRA, MOH)
- ✓ Prepares dossiers, compliance documentation
- ✓ Communicates with relevant authorities

3.6. Supply Chain & Logistics

- ✓ Procurement of raw materials, capsules, packaging
- ✓ Warehouse & stock management
- ✓ Delivery/distribution planning

3.8. Finance & Accounting

- ✓ Manages budgeting, salaries, audits
- ✓ Prepares financial statements
- ✓ Handles L/Cs, T/Ts, and bank relations

3.9. Marketing and Sales

- ✓ Develops branding & promotion strategy
- ✓ Manages media, partnerships, and advertising
- ✓ Supports product launch

3.10. Human Resources (HR)

- ✓ Staff recruitment, onboarding, and training
- ✓ Oversees compliance with labor
- ✓ Evaluates performance & HR policies

3.11. Facility Engineering & Maintenance

- ✓ Maintains cleanroom, HVAC, utilities
- ✓ Ensures GMP-compliant infrastructure
- ✓ Coordinates equipment calibration/repairs

3.12. Risk Management

- ✓ Identifies. Accesses and monitors risks that could impact operations, finance or compliance
- ✓ Develops mitigation strategies and contingency plans to minimize potential losses
- ✓ Ensures integration of risk controls into detail operations and strategic planning

3.13. IT & Data Systems

- ✓ Manages data integrity, cyber security, LIMS/ERP systems
- ✓ Supports automation and traceability
- ✓ Digital compliance (e.g., 21 CFR Part 11)

3.14. Licensing and Regulatory Compliance:

AGF-PHYTOMED INDUSTRIES is subjected to the following regulatory licensing:

- Business Application & Registration –PACRA
- Investment Licence- ZDA
- Environmental Impact Assessment (EIA) Certificate- ZEMA
- Screening EPB Report Certificate- ZEMA
- Waste Management Licence- ZEMA
- Site Approval Certificate- ZMRA
- Product Registration- ZMRA
- GMP Inspection- ZAMRA
- Manufacturing Licence- ZAMRA
- Investor Permit- IMMIGRATION
- Product Testing & Application- ZABS
- Factory Inspection- ZABS
- Certification Mark- ZABS
- Premises Hygiene Clearance Certificate- Local Health Department (Munispal/District Health Office)
- Food Safety Inspection/Compliance Certificate- Local Health Department (Munispal /District Health Office)
- Food Handling/Safety Certificate- Local Health Department (Munispal /District Health Office)
- Occupational Health Risk Assessment Certificate- Local Health Department (Munispal/District)
- Local Health Department (Munispal /District Health Office)- Annual Health Compliance Verification
- Land Title Deeds- Headmen, Chief, Chiebombo District, Province &Ministry of Land
- ISO 9001-Quality Management System- Initial Certification
- ISO 9001-Quality Management System- Annual Surveillance
- ISO 22000-Food Safety
- ISO 14001- Environmental Gouvernance and Sustainability- Initial Certification
- ISO 14001- Environmental Gouvernance and Sustainability- Annual Surveillance
- ISO 45001- Occupational Health & Safety- Initial Certification
- ISO 45001- Occupational Health & Safety- Annual Surveillance Audits
- ISO 14644- Cleanroom Qualification ISO 5/Room x 10
- ISO 14644- Cleanroom Annual Re-certification Audits ISO 5/Room x 10

Page 7: AGF-PHYTOMED INDUSTRIES Facility

FACILITY LAYOUT PLAN REPORT FOR AGF PHYTOMED INDUSTRIES

	<i>L = Length</i>		2F = 2 floors		
	<i>W = Width</i>		HS = high shelves		
	<i>A = Area</i>				
	<i>H = Floor-to-ceiling height</i>				
	<i>For irregular shapes, only A is mentioned</i>				

A. INDUSTRIAL BULDING							
				L(m)	W(m)	A(m2)	H(m)
I	RECEPTION	Reception		5	4	20.0	4
		Waiting seats		5	2.5	12.5	4
		Staff gathering area				150.0	4
II	OFFICES	Self-contained and shared Offices				370.0	4
		Staff Meeting room		11	8	88.0	4
		Toilets				62.0	4
III	DOWNLOADING AREA			43	4	172.0	6
IV	RAW MATERIALS	Sample storage	HS	9	5	45.0	6
		Wraps shelves keeping	HS	6	5	30.0	6
		Reagent chemical storage	2F	15	10	150.0	6
		Raw material storage	2F	15	10	150.0	6
		Dry Raw material storage	2F	15	10	150.0	6
		Quarantine storage	HS	5	10	50.0	6
		Equipment storage	HS	3	10	30.0	6
		Property store	HS	9	5	45.0	6
		Gardener store	HS	5	5	25.0	6
V	POST HARVEST & PRE-PROCESSING	Extracts		40	10	400.0	8
		Soymilk powder					
		Liquid soy milk					
		Dried tofu		40	30	1,200.0	8
		Spirulina algae		40	10	400.0	8
		Fruit juices		40	10	400.0	8
		Reserved hall		40	10	400.0	8
VI	PROCESSING AND PACKAGING	Extracts		70	10	700.0	8
		Soymilk powder					
		Liquid soy milk					
		Dried tofu		70	30	2,100.0	8
		Spirulina algae		70	10	700.0	8
		Fruit juices		70	10	700.0	8
		Reserved hall		70	10	700.0	8
VII	AISLE	All around the processing halls			2.5		8
VIII	FINISHED PRODUCTS	Industrial cold room	HS	17.5	10	175.0	6
		Finished product store	2F	13	10	130.0	6
		Warehouse	2F	15	10	150.0	6
		Quarantine storage	HS	5	10	50.0	6
		Waiting seats		5	5	25.0	4

IX	INDUSTRIAL COLD ROOM			17.5	10	175.0	6
X	PACKAGING AND UPLOADING			53.5	4	214.0	6
XI	CHANGING ROOMS	Changing rooms 1		17.5	13	227.5	4
		Changing rooms 2		13	12.5	162.5	4
XII	LABORATORIES	Analytical Chemistry		10	5	50.0	6
		Microbiology		10	5	50.0	6
		Formulation /R&D		10	5	50.0	6
		Stability & Shelf-Life		10	5	50.0	6
		Raw materials/Plant identification		10	5	50.0	6
		Labo storeroom					
XIII	LABORATORY ANIMAL BREEDING FACILITY			20	15	300.0	6
XIV	UTILITIES AND TECHNICALS	HVAC		7	6.5	45.5	4
		Water system		7	5.5	38.5	4
		Compressors		7	5.5	38.5	4
		Power back up		7	5.5	38.5	4
XV	WASTE AREA	Waste storage		10	10	100.0	4
		Waste management zone		20	10	200.0	
		Including Waste segregation bins					
XVI	PARKING	Staff and visitors parking		78	5	390.0	
		Trucks parking 1		20	8	160.0	
		Trucks parking 2		19	8	152.0	
		TOTAL BUILT AREA (The building)				13,640.0	
FENCING				200	150	30,000.0	
PLUS THE SITE CIRCULATION ROADS							
				h=3.2 (floor 1)			
				h=3.5 (floor 2)			
XVII	STAFF RESIDENTIAL UNITS	Seating room		4	2.7	10.8	
		Dining room + kitchen		2.7	2.5	6.8	
		Toilet		2	1.5	3.0	
		Bedroom		3.75	2.5	9.4	
		Veranda		5	1.5	7.5	
		Restaurant + Kitchen		8	5	40.0	

		Supermarket		6	3	18.0	
		Coffee space		5.5	3	16.5	
		Parking		50	5	250.0	
		TOTAL BUILT AREA (The building)				620.0	
PLUS, THE SITE CIRCULATION ROADS							
	FENCING			70	40	2800	
XVIII	GARAGE	Reception		5	5	25.0	4
		Spare parts and storage		8	5	40.0	4
		Utility and technical room		5	5	25.0	4
		Mechanical workshop		10	8.5	85.0	4
		Tire and wheel services		10	10	100.0	4
		Quick service area		10	10	100.0	4
		Car wash service		40	20	800.0	
		Toilets		5	1.5	7.5	4
		TOTAL BUILT AREA		40	30	1,200.0	
PLUS, THE SITE ACCES ROADS							
	FENCING			40	30	1,200.0	

Page 8: Our Product Categories

Product 1: GMP-compliant plant-derived **Crude and Purified Vacuum Dried Extracts** intended for processing standardized TCAMs, Nutraceuticals/Food Supplements, Phytomedicines and Phytopharmaceuticals (Capsules, tablets, concentrated syrups, liquid extracts and powdered extracts, micronized powders, etc.) targeting acute and chronic diseases, including diabetes, high blood pressure, cancer and benign tumors, cerebro-cardio-vascular diseases.

Product 2: Standardized & nutrient-dense **Fortified Soybean Milk Powder** – Rich in protein and essential micronutrients, catering to health-conscious consumers;

Product 3: Standardized & nutrient-dense **Fortified Liquid Soymilk** – Plant-based complete protein alternatives, powerhouse of essential micronutrients;

Product 4: Standardized & nutrient-dense **Fortified and Vacuum-dried Tofu**– Plant-based complete protein alternatives, powerhouse of essential micronutrients;

Product 5: Clean-label **Fruit Juices** – Locally sourced nutrient-dense fruits processed into nutritious beverages free from added sugar or chemicals;

Product 6: Standardized **Fortified Spirulina Algae** Supplements (Tablets, capsules and micronized powder)–Power house of complete proteins and essential micronutrients with global demand;

Product 7: High-Nutrient GMP-Compliant **High-nutrient Bio-Organic Fertilizer** made from fermented cow's fresh milk, cow's and pig's + dung and urines, lactic

bacteria stains and lab strongest stabilizer, completely free from added chemicals detrimental to the environment and consumers, made essentially from fermented (Bioconversion) pig dung & urine, cow's milk, lactic acid bacteria (LAB), molasses and cryoprotectant (Tetrahalose=Lactic bacteria protectant).

Page 9: Raw Material Acquisition

AGF gets raw materials from the following channels:

- (1) Contractual certified supplying companies of plant raw materials (e. g. India & China), during the first five years of operation;
- (2) AGF's Commercial Organic Farming of 90ha-farmland, with expansion plans to over 2000 hectares.

Page 10: Industrial Technology & Processing Systems

Category 1: CRUDE AND PURIFIED EXTRACTS PROCESSING WORKFLOW:

- TCAMs (Traditional, Complementary, and Alternative Medicines)
- Nutraceuticals/Food Supplements
- Phytomedicines
- Phytopharmaceuticals

With our customized, fully integrated and automated Turnkey Ultrasound-Assisted Extraction (UAE), Capsule Production, Tableting and Packaging Line System:

- Step 1: Papaya Leaf Powder reception and inspection
- Step 2: Degreasing
- Step 3: Extraction of target bioactive compounds using specific organic solvent
- Step 4: Concentration
- Step 5: Chromatographic purification
- Step 6: Vacuum drying
- Step 7: Micronization and ultrasonic vibratory sieving
- Step 8: Anti-cracking agent aseptic incorporation
- Step 9: Initial profiling and standardization
- Step 10: Capsule processing/tableting
- Step 11: Bottle filling, capping and sealing
- Step 12: Instructional labeling
- Step 13: Cartoning and leaflet inserting
- Step 14: Case packaging
- Step 15: Post-packaging High Pressure Processing (HPP) Optimization (Non-thermal sterilization)
- Step 16: Production profiling and standardization
- Step 17: Inspection, palletizing and dispatch

Category 2: INSTANT FORTIFIED SOYMILK POWDER PROCESSING WORKFLOW:

With our customized, fully integrated and automated Turnkey Line System designed specifically for Fortified Liquid Soybean Milk, Soymilk Powder and Fortified Tofu:

- Step 1: Raw materials (Dry soybeans) reception and inspection
- Step 2: Cleaning, grading and sorting
- Step 3: Dehulling and soaking
- Step 4: Wet grinding
- Step 5: Slurry heating
- Step 6: Filtration (Okra separation)
- Step 7: Enzymatic deactivation
- Step 8: Homogenization
- Step 9: Initial profiling and standardization
- Step 10: Ultra-High Pressure Processing (HPP) Optimization (Post-harvest) at 40°C (Temperature) and 400-600 MPa (Pressure). for microbial control before the processing takes place
- Step 11: Initial profiling and Initial standardization
- Step 12: Vacuum drying into fine powder
- Step 13: Jet milling (Micronizer) for micronization into ultrafine powder (1-10 micrometers (Increased surface area, better homogeneity, improved stability, enhanced absorption)
- Step 14: Ultrasonic vibratory sieving (Vibrating Screen Classifier) and cooling
- Step 15: Anti-caking agent incorporation (e. g. silicon dioxide) with High-Shear mixer
- Step 16: Aseptic micronutrient fortification (*Vitamins A, B-complex, C, D and E, Omega-3, zinc, selenium, magnesium, cysteine*, etc., natural flavoring agents)
- Step 17: Final product profiling and micronutrient standardization
- Step 18: Aseptic filling into 250g, 500g and 1kg tinplate milk cans with EOE & plastic overcap
- Step 19: Nitrogen flashing and can sealing
- Step 20: Instructional labeling & batch coding
- Step 11: Post-packaging sterilization with Ultra-High Pressure Processing (HPP) Optimization for high-risk products, to maximize safety with regard to pathogen content of the finished products
- Step 22: Case packaging (6 cans/box)
- Step 23: Plastic crate packaging
- Step 24: Inspection, palletizing and dispatch

Category 3: FORTIFIED LIQUID SOYMILK PROCESSING WORKFLOW:

With our customized, fully integrated and automated Turnkey Line System designed specifically for Fortified Liquid Soybean Milk, Soymilk Powder and Fortified Tofu:

- Step 1: Raw materials (Dry soybeans) reception and inspection
- Step 2: Cleaning, grading and sorting
- Step 3: Dehulling and soaking
- Step 4: Wet grinding
- Step 5: Slurry heating
- Step 6: Filtration (Okra separation)
- Step 7: Enzymatic deactivation
- Step 8: Homogenization
- Step 9: Initial profiling and standardization
- Step 10: High Pressure Processing (HPP) Optimization (Post-harvest) at 40°C (Temperature) and

400-600 MPa (Pressure). for microbial control before the processing takes place

- Step11: Aseptic micronutrient fortification (*Vitamins A, B-complex, C, D and E, Omega-3, zinc, selenium, magnesium, cysteine, natural flavoring agents*)
- Step 12: Initial product profiling and micronutrient standardization
- Step 13: Aseptic filling (e. g. 500ml or 1L in Tetra Brik Aseptic Cartons)
- Step 14: Carton sealing
- Step 15: Date coding and inspection
- Step 16: Inspection, palletizing and dispatch
- Step 17: Case packaging (Corrugated boxes)/Plastic crate packaging
- Step 18: Inspection, palletizing and dispatch

Category 4: FORTIFIED VACUUM DRIED TOFU PROCESSING WORKFLOW:

With our customized, fully integrated and automated Turnkey Line System designed specifically for Fortified Liquid Soybean Milk, Soymilk Powder and Fortified Tofu:

- Step 1: Raw materials (Dry soybeans) reception and inspection
- Step 2: Cleaning, grading and sorting
- Step 3: Dehulling and soaking
- Step 4: Wet grinding
- Step 5: Slurry heating
- Step 6: Filtration (Okra separation)
- Step 7: Coagulation
- Step 8: Mechanized filtration and pressing system (Initial pressing) = <60 moisture
- Step 9: Aseptic micronutrient fortification and marination (*Vitamins A, B-complex, C, D and E, Omega-3, zinc, selenium magnesium cysteine,, etc., marination, natural flavoring agents*)
- Step 10: Cutting and portioning
- Step 11: Initial profiling and initial standardization
- Step 12: Freeze drying
- Step 13: Final product profiling and micronutrient standardization
- Step 14: Packaging into 250g, 500g and 1kg food-grade cartons with EOE & plastic overcap
- Step 15: Modified atmosphere packaging
- Step 16: Plastic crate packaging
- Step 17: Corrugated box case packaging (6 cartons/box)
- Step 18: Post-packaging sterilization with Ultra-High Pressure Processing (HPP) Optimization for high-risk products, to maximize safety with regard to pathogen content of the finished products
- Step 19: Inspection, palletizing and dispatch

Category 5: FORTIFIED SPIRULINA ALGAE SUPPLEMENT PROCESSING WORKFLOW:

With our customized, fully integrated and automated Turnkey Line System designed specifically for spirulina algae production and harvesting, powder, capsule, tablet and processing as well and packaging:

- Step 1: Inoculum Preparation (*Pure Spirulina culture under sterile conditions Grown in pre-culture photobioreactors (controlled light, temp, CO₂)*)
- Step 2: Mass Cultivation in closed photobioreactors
- Step 3: Harvesting (*Continuous flow disc-stack centrifugation or tangential flow filtration (TFF), spirulina biomass separated from culture medium*)

- Step 4: Washing & Dewatering (*Triple washing with sterile water to remove salts & residues, fewatered to paste (70–80% moisture)*)
- Step 5: Vacuum drying
- Step 6: High Pressure Processing (HPP) Optimization (Post-harvest) at 40°C (Temperature) and 400-600 MPa (Pressure) for microbial control before the processing takes place
- Step 7: Jet milling for micronization
- Step 8: Aseptic micronutrient fortification
- Step 9: Blended powder in food-grade tinplate cans with EOE & plastic overcap (Powder as final product)
- Step 10: Encapsulation/Tableting (Capsules and tablets as final products)
- Step 11: Bottling
- Step 12: Capping and sealing
- Step 13: Instructional labeling
- Step 14: Cartoning
- Step 15: Post-packaging sterilization with High Pressure Processing (HPP) Optimization for high-risk products, to maximize safety with regard to pathogen content of the finished products
- Step 16: Leaflet inserting
- Step 17: Case packaging (6 bottles/box each)
- Step 18: Inspection, palletizing and dispatch

Category 6: FORTIFIED COLD-PRESSED FRUIT JUICES PROCESSING WORKFLOW:

With our customized, fully integrated and automated Turnkey Cold Press Line System designed specifically for juice processing and packaging:

- Step 1: Raw materials reception and inspection
- Step 2: Fruit washing and sorting
- Step 3: Cold pressing
- Step 4: Filtration and clarification
- Step 5: Initial profiling and initial standardization
- Step 6: Pasteurization
- Step 7: Aseptic micronutrient fortification (*Vitamins A, B-complex, C, D and E, zinc, selenium magnesium cysteine, etc. natural flavoring agents*)
- Step 8: Homonization
- Step 9: Final product profiling and micronutrient standardization
- Step 10: Aseptic bottle filling/Cartoning (*PET bottles, glass bottles, food-grade aseptic cartons*)
- Step 11: Capping
- Step 12: Instructional labeling
- Step 13: Post-packaging sterilization with High Pressure Processing (HPP) Optimization for high-risk products, to maximize safety with regard to pathogen content of the finished products
- Step 14: Case & plastic crate packaging
- Step 15: Inspection, palletizing and dispatch

Category 7: HIGH-NUTRIANT ORGANIC FERTILIZER PROCESSING WORKFLOW:

A) Liquid Concentrate:

With Anaerobic Fermentation Bioreactor:

- Step1: Collection, inspection and qualification of organic substrates (cow & pig dung/urines, cow's fresh milk & LAB strains)

Step 2: Raw material screening, homogenization and pre-treatment
 Step 3: Controlled anaerobic (Bioreactor) lactic fermentation
 Step 4: In-process monitoring (pH, temperature, CPU count, organic acid profile)
 Step 5: Microfiltration for initial microbial control
 Step 6: Vacuum evaporation (Concentration) at the reduced pressure of 40°C for liquid concentrate optimization
 Step 7: Incorporation of Trehalose for LAB stabilization
 Step 8: High-pressure processing (HPP) Optimization (40°C)
 Step 9: Profiling and standardization
 Step 10: Aseptic bottle filling and capping
 Step 11: Instructional labeling and batch coding
 Step 12: Case packaging
 Step 13: Final inspection, palletizing and dispatch

B) Micronized Powder:

Step 1: Collection, inspection and qualification of organic substrates (cow & pig dung/urines, cow's fresh milk & LAB strains)
 Step 2: Raw material screening, homogenization and pre-treatment
 Step 3: Controlled anaerobic (Bioreactor) lactic fermentation
 Step 4: In-process monitoring (pH, temperature, CPU count, organic acid profile)
 Step 5: Microfiltration for initial microbial control
 Step 6: Vacuum evaporation (Concentration) at the reduced pressure of 40°C into paste form
 Step 7: Vacuum drying
 Step 8: Micronization
 Step 9: Incorporation of Trehalose for LAB stabilization
 Step 10: High-pressure processing (HPP) Optimization (40°C)
 Step 11: Profiling and standardization
 Step 12: Aseptic packaging into moisture-proof aluminum 0.5kg, 1, 5 & 10kg sachets and 1-5kg HDPE bottles
 Step 13: Nitrogen flashing
 Step 14: Bottle capping and sealing
 Step 15: Instructional labeling and batch coding
 Step 16: Case packaging
 Step 17: Final inspection, palletizing and dispatch

Page 11: Market & Industry Outlook

➤ **Domestic Market Viability:**

- ✓ Growing non-communicable disease prevalence and increasing consumer awareness of preventive healthcare are driving demand for nutraceuticals, fortified functional foods, and phytomedicines. Localized production supports affordability and import substitution.

➤ **Regional Market Opportunity:**

- ✓ COMESA, EAC and SADC markets present scalable demand for standardized botanical extracts and evidence-based natural health products, with limited GMP-aligned production facilities currently operating at scale.

➤ **International Export Potential:**

- ✓ Global demand for clean-label botanical extracts and clinically validated phytomedicines continues to expand. AGF's structured certification roadmap (e. g. ISO 9001, ISO 22000, GMP compliance) supports long-term access to regulated export markets.

Page 12: Digital Growth & Market Expansion Platform

AI-Powered, Multi-Channel, Community-Driven Market Growth

1) Strategic Overview

AGF's Digital Growth & Market Expansion Platform centralizes all strategic digital marketing channels, including existing and ever expanding educational WhatsApp groups, Telegram, Facebook, Instagram, TikTok, YouTube, email, and website—into a single coordinated system using HubSpot Marketing Hub. Integrated with ChatGPT AI, this platform automates daily content, advertisements, and community engagement, ensuring scalable, measurable, and professional market expansion.

Investor Advantage: Transparent KPIs, real-time analytics, and predictable growth across domestic, regional, and international markets.

2) Community & Expert Engagement

235+ WhatsApp groups, 211,500+ participants, expanding daily

Key Partner: NUFOTEC BURUNDI (EAC)

Expert-Led Content: Registered nutritionists, dieticians, and medical professionals provide daily guidance on presentational advices and guidance on:

- Acute diseases: e. g. cough, diarrhea, respiratory infections
- Chronic conditions: cancer, diabetes, cardiovascular disease, obesity, child malnutrition
- Preventive nutrition, home remedies, and organic farming
- Communities act as active marketing and educational channels, converting engagement into product adoption and economic development.

3) ChatGPT AI Services

Role: Drives all automated content and advertising across AGF's strategic digital marketing channel network

Automated Daily Content: Health tips, preventive nutrition, product updates, and home remedies distributed across all our platforms

Automated Advertisements: Targeted domestic, regional, and international campaigns

Real-Time Community Interaction: 24/7 AI-powered assistance and feedback capture

Analytics & Reporting: Tracks engagement, adoption, and ESG impact via real-time dashboards

Impact: ChatGPT ensures continuous, coordinated, and measurable outreach, transforming communities into active market channels.

4) Market Expansion Impact

- Fast adoption and brand recognition
- Regional (EAC, COMESA, SADC)
- AI-optimized content for cross-border engagement
- Easy international outreach
- Automated campaigns reach global distributors and partners

5) Investor & Donor Highlights

- Proven traction: 235+ WhatsApp groups, 211,500+ participants
- AI-powered HubSpot + ChatGPT automation ensures consistent engagement and daily updates
- Expert-led guidance builds trust and adoption
- Real-time dashboards provide transparent KPIs and ESG reporting
- Single coordinated system enables scalable, predictable growth

Footer / Tagline:

Powered by AGF company and HubSpot integrated with ChatGPT AI, AGF's platform delivers automated daily content, expert guidance, and community-driven product promotion, transforming communities into active growth channels for health, economic development, and global market expansion

Page 13: Digital Health Consultation

- Electronic intake forms
- Telephonic Health Consultation

- Virtual Medical (MD) Consultation
- Live Video Teleconsultation
- Preventive Wellness Advice for:
 - ✓ Shop/E-Commerce
 - ✓ Product catalog
 - ✓ Retail Orders
 - ✓ Bulk Orders
 - ✓ Distribution Access
 - ✓ Secure Payment Portal

Page 14: Phased Investment Projection

African Green Farmers Limited (AGF)– AGF-PHYTOMED INDUSTRIES FACILITY– seeks to mobilize USD 40+ million in phased seed capital (2026–2029) dedicated not to commercial farming, but to construction, commissioning, and operationalization of a GMP-compliant phytomedicine and nutraceutical processing facility.

During the first 5 years of operation, AGF will not allocate seed capital to commercial farming activities. All raw botanical, Spirulina and organic fertilizer inputs will be secured through structured external contractual supply agreements.

This model significantly reduces early-stage agricultural risk and accelerates industrial revenue generation.

1) Seed Capital Allocation Strategy (Years 1–5)

Seed capital will be allocated strictly to the initial industrial, regulatory, and commercialization infrastructure.

A. Investing in Facility Construction & Industrial Infrastructure

- Construction of the AGF-PHYTOMED INDUSTRIES processing facility
- Pre-built GMP cleanrooms for critical processing zones
- HVAC systems with controlled pressure gradients
- Clean steam and purified water systems
- Waste management and solvent recovery systems
- Hybrid power systems (solar + backup generation)
- Fire suppression and safety compliance systems
- Staff Residential Units

B. Investing in Processing & Production Equipment

A fully automated and integrated high-throughput Turnkey *Ultrasound-Assisted Extraction (UAE) Line System* integrating (In a single line system) essentially post-harvest processing of plant raw materials, validated solvent-based extraction, concentration, chromatographic

purification, vacuum drying, micronization and ultrasonic vibratory sieving, anti-cracking agent aseptic incorporation, capsule and tablet production, bottle filling, capping and sealing, cartooning, instructional labeling and leaflet inserting, post-packaging High Pressure Processing (HPP) Optimization (Non-thermal sterilization) and case packaging.

This is a GMP-Compliant manufacturing efficiency within a single continuous end-to-end line system, approximately 60m x 5m in size. The line system manufactures any target extracts from any medicinal plants based on our need. This is the only processing line system to be secured using the seed capital.

The rest of manufacturing line systems such as the Soybean Product (Fortified Liquid Soymilk, Soymilk Powder and Vacuum-dried Tofu) Integrated Manufacturing Line System, Spirulina Algae Supplement Integrated Turnkey Processing Line System, Fruit Juice Integrated Turnkey Processing Line System, and Organic Fertilizer Integrated Turnkey Processing Line System will be secured and utilized using the facility's income-generating activities, mainly sales.

C. Investing in Laboratory & Quality Control Infrastructure

➤ Analytical laboratory and scientific research laboratory equipment, including UHPLC-MS/MS, GC-MS/MS 8000, HPAAS, AH SN600 Laser Scanning Microscope, RT-qPCR and High-Parameter Digital Flow Cytometer for:

- 1) **Microbiology laboratory** for testing microorganisms and biological contamination of raw materials and finished products
- 2) **Analytical laboratory (QC Lab)** that performs testing for product profiling and standardization
- 3) **Preclinical laboratory** performing product toxicology, safety, and pharmacokinetics
- 4) **GLP (Good Laboratory Practice) laboratory** that generates non-clinical data before human trials
- 5) **Clinical research laboratory** that analyses clinical study samples

D. Investing in Regulatory, Licensing & Certification

Regulatory and licensing in which the seed capital will be utilized include the following:

- Business Application & Registration –PACRA
- Investment Licence- ZDA
- Environmental Impact Assessment (EIA) Certificate- ZEMA
- Screening EPB Report Certificate- ZEMA
- Waste Management Licence- ZEMA
- Site Approval Certificate- ZMRA
- Product Registration- ZMRA
- GMP Inspection- ZAMRA
- Manufacturing Licence- ZAMRA

- Investor Permit- IMMIGRATION
- Product Testing & Application- ZABS
- Factory Inspection- ZABS
- Certification Mark- ZABS
- Premises Hygiene Clearance Certificate- Local Health Department (Munispal/District Health Office)
- Food Safety Inspection/Compliance Certificate- Local Health Department (Munispal /District Health Office)
- Food Handling/Safety Certificate- Local Health Department (Munispal /District Health Office)
- Occupational Health Risk Assessment Certificate- Local Health Department (Munispal/District
- Local Health Department (Munispal /District Health Office)- Annual Health Compliance Verification
- Land Title Deeds- Headmen, Chief, Chiebombo District, Province &Ministry of Land
- ISO 9001-Quality Management System- Initial Certification
- ISO 9001-Quality Management System- Annual Surveillance
- ISO 22000-Food Safety
- ISO 14001- Environmental Gouvernance and Sustainability- Initial Certification
- ISO 14001- Environmental Gouvernance and Sustainability- Annual Surveillance
- ISO 45001- Occupational Health & Safety- Initial Certification
- ISO 45001- Occupational Health & Safety- Annual Surveillance Audits
- ISO 14644- Cleanroom Qualification ISO 5/Room x 10
- ISO 14644- Cleanroom Annual Re-certification Audits ISO 5/Room x 10

E. Investing in Operationalization & Market Deployment

- Salaries for skilled staff recruitment and training
- Salaries for executive management
- Rent of headquarters, domestic health shops and regional distribution outlets
- Distribution fleet acquisition
- Executive coordination vehicle
- ERP and digital traceability systems
- Branding, packaging design, and market launch
- Insurance (facility, liability, professional indemnity)
- Security systems and facility access control

2) Raw Material (Years 1–5)

During the first operational phase:

- Raw medicinal plant materials sourced through contractual supplier agreements
- Spirulina biomass procured through structured external production partnerships
- Multi-supplier diversification to reduce dependency risk
- Quality specifications enforced through pre-supply laboratory validation

This strategy ensures:

- Lower early-stage capital exposure
- Reduced climate and agricultural production risk
- Faster time-to-market
- Immediate focus on value-added processing

3) Revenue Generation Model (Years 1–5)

Commercial farming expansion and additional equipment acquisition will be financed through internally generated revenues from:

- Standardized botanical extracts (TCAMs, Nutraceuticals/Food Supplements, Phytomedicines and Phytopharmaceuticals—Tablets, capsules, micronized powders, liquid concentrates)
- High-nutrient Organic Fertilizers (Micronized powders and liquid concentrates)
- Contract manufacturing services (third-party processing)
- Private label production
- Export of semi-processed phytochemical ingredients
- Laboratory analytical services
- Licensing and formulation partnerships

Revenue reinvestment will finance mainly:

- Commercial organic cultivation clusters of target medicinal plants, functional crops, fruits...
- Acquisition of additional high-capacity processing lines
- Vertical integration of priority medicinal crops
- Regional scaling of distribution infrastructure

4) Strategic Rationale of This Model

This phased approach:

- Reduces agricultural CAPEX exposure in early years
- Converts seed capital directly into revenue-generating industrial assets of primary priorities
- Accelerates EBITDA stabilization
- De-risks investor capital
- Strengthens bankability
- Preserves long-term vertical integration potential
- Investors finance a processing and value-add industrial facility first — not raw agricultural risk.

5) Investor Confidence Positioning

This structure demonstrates that AGF:

- Prioritizes industrial cash-flow generation
- Minimizes early-stage climate exposure
- Maintains disciplined capital utilization
- Uses retained earnings to expand vertically
- Aligns with private equity and DFI risk expectations

Page 15: Our Investor & Partner Commitment

(Building Trust Through Transparency, Accountability, and Strategic Growth)

1) Our Commitment to Investors, Donors & Financial Brokers

At AGF, we are dedicated to maximizing impact while ensuring financial integrity. Our approach combines:

- Non-fund Diversion Policy: Legally binding restricted use of proceeds covenant
- Transparency: Clear reporting of funds, expenditures, and project outcomes
- Accountability: Governance structures and audit-ready processes
- Sustainability: Long-term strategies for profitable, socially responsible, and ESG-aligned growth

Our goal is to ensure every investment and donation directly contributes to market expansion, sustainable community development and health impact.

2) Financial Governance and Oversight

AGF adheres to international best practices in financial management, including:

- Independent audits and financial reviews
- Robust internal controls to safeguard assets
- Board-level oversight of all financial decisions
- Compliance with local and international regulations and reporting standards
- Investors and donors can trust that funds are managed efficiently, ethically, and in full compliance.

3) Investment Opportunities

AGF offers strategically structured investment opportunities designed to generate measurable returns while supporting health, nutrition, sustainable economic development and sustainable agriculture initiatives:

- Seed and growth-stage funding for AGF-PHYTOMED INDUSTRIES and digital platforms
- Co-investment opportunities with regional and international partners

- ESG-aligned projects focusing on preventive nutrition, functional foods, and organic farming of target medicinal plants, functional nutrient-dense crops, fruits and vegetables serving as raw materials for our manufacturing facility
- Investments are linked to clear KPIs and market expansion metrics, ensuring measurable impact and visibility.

4) Donor Engagement and Impact

Donors are a critical part of AGF's mission. We provide:

- Regular impact reports highlighting social, environmental, and health outcomes
- Transparent tracking of funds toward community programs, research, and digital outreach
- Recognition of donors in line with preferences and contribution scale
- Donors can see how every contribution translates into real-world health, economic, gender-sensitive job creation and social impact.

5) Financial Reporting and Transparency

AGF prioritizes real-time financial visibility:

- Quarterly and annual financial statements
- Online dashboards with project-wise and program-wise fund allocation
- Key Performance Indicator (KPI) tracking linked to market growth, community engagement, and ESG performance

This ensures investors and donors can monitor the impact of their support anytime.

6) Why Partner With AGF?

- Proven track record: 235+ WhatsApp groups, 211,500+ participants, growing daily
- AI-driven digital marketing and community engagement for measurable reach
- Robust governance, compliance, and audit-ready processes
- Clear investment structure with ESG-aligned, high-impact projects
- Dedicated investor, donor & broker relations team providing personalized support and reporting

Footer / Call-to-Action:

- Join AGF in driving innovative health solutions, commercial agriculture, preventive nutrition and sustainable socio-economic development.
- Invest, Donate or Make Brokerage Agreement with us today – Partner with us to create measurable impact across Africa and beyond

At AGF:

- We view *Capital Introduction Brokers* as strategic partners and protect commercial interests
- We honor and strictly adhere to Broker Agreement bearing the Public Notary's endorsement with absolute integrity and transparency
- Commissions agreed upon in our bilateral agreement are paid as promptly as the successful funds are secured/disbursed
- No upfront commission fees accepted

Page 17: Risk Analysis & Mitigation Strategies

AGF applies a comprehensive Enterprise Risk Management (ERM) framework tailored to African business realities below:

No	RISKS	MITIGATION STRATEGIES
1	Drought, erratic rainfall, water scarcity, and logistics challenges → reduced yields	Drip/sprinkler irrigation, drought-tolerant varieties, integrated pest management, climate monitoring and geographic diversification
2	ESG and Community Risks: Land disputes or reputational issues	Land title deeds, community inclusion programs, outgrower contracts, ESG reporting.
3	Load-shedding affecting negatively the processing and the cold chain	Solar/diesel hybrid power systems, backup industrial generator, energy-efficient processing
4	Fluctuating prices, foreign exchange	Diversified products (Tablets, capsules, powders...), contract farming & forward sales agreements
5	Soil depletion, biodiversity impact	Agroforestry, organic fertilizers, soil testing, sustainable land management
6	Quality loss, spoilage, inefficiencies	GMP-compliant processing, cold storage, optimized logistics, HACCP protocols
7	Worker illness, labor shortage	Occupational health programs, sanitation, training, insurance coverage
8	Crop, machinery, or input theft and vandalism	▪ Security wall fence ▪ Security wire fence surrounding the whole farmland ▪ Security guards ▪ Perimeter security, CCTV cameras, community engagement, access controls
9	Unexpected losses (climate, market, fire)	Crop and asset insurance, contingency funds, prudent financial planning

10	Macroeconomic and Foreign Exchange Risks: Currency fluctuations, inflation, and import constraints	USD-denominated exports, multi-currency accounts, and natural hedging
11	Political and Regulatory Risks: Policy shifts, licensing delays, and taxation changes	Proactive engagement with regulators, legal advisory, and diversification across markets
12	Supply Chain Risks: Variability in raw materials and potential cooperative defaults	contractual suppliers, local cooperatives, quality pre-testing, and multiple sourcing options
13	Governance and Capital Risks: Misallocation or weak controls	milestone-based disbursement, segregated accounts, IFRS-aligned audits, and investor oversight
14	Market Access Risks: Payment delays and counterfeit products	Pre-offtake agreements, digital traceability, and diversified distribution channels.

Page 18: Strategic Partnerships

Financial, Technical and Scientific Partners

AGF-PHYTOMED INDUSTRIES FACILITY collaborates with a carefully structured network of domestic, regional, and international partners. These partnerships integrate scientific research, clinical validation, manufacturing expertise, regulatory guidance, and financial support to ensure that all natural medicines we produce are safe, standardized, and globally credible.

1. Research & Scientific Validation

Purpose: Validate plant-based compounds for chronic diseases including cancer, diabetes, and cardiovascular conditions.

Domestic: University of Zambia – preclinical research, biomarker identification, and translational studies.

Regional: University of Cape Town – pharmacology studies and chronic disease modeling.

International: Harvard Medical School, University of Oxford, Karolinska Institutet – mechanism validation, translational research, and joint intellectual property development.

2. Clinical Trials & Healthcare Networks

Purpose: Conduct human studies to demonstrate safety and efficacy.

Domestic: Lusaka hospitals and NCD clinics – patient recruitment and clinical monitoring.

Regional: Regional referral hospitals – coordination and management of clinical studies.

International: MD Anderson Cancer Center, Mayo Clinic – oncology and metabolic trial collaboration, statistical analysis, and GCP compliance.

3. Regulatory & Market Authorization

Purpose: Ensure compliance with domestic and international standards.

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